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Eco Friendly Robot

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Eco Friendly Robot
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ABSTRACT

In response to the accelerating global waste crisis and the demand for sustainable, cost-effective recycling solutions, we present a real-time waste-classification and sorting system powered by deep learning. A fine-tuned ResNet-18 Convolutional Neural Network (CNN) executes on a GPU-equipped laptop, analysing live video from a smartphone camera and distinguishing recyclable from non-recyclable items. Whenever three of the last five frames exceed a 70 % non-recyclable confidence threshold, the laptop transmits a one-byte TCP command over Ethernet to a Raspberry Pi 4, which then drives GPIO 17 HIGH for three seconds to energise a relay-controlled pneumatic piston that diverts the item on a conveyor belt.

The model was trained on a custom dataset of 1 801 images (1 013 recyclable, 788 non-recyclable) captured under varied illumination and angles. Online augmentation—random rotation, horizontal flip, colour jitter and Gaussian blur—was applied at every epoch, and 3-fold cross-validation with class-balanced loss mitigated sampling bias. On the held-out test set the network achieved 99.3 % accuracy, 0.993 F1 and 0.998 ROC-AUC, while end-to-end latency from camera capture to piston actuation remained below 90 ms. The hybrid laptop–Pi topology therefore delivers industrial-grade precision and responsiveness without requiring expensive edge GPUs, making it suitable for small-scale recycling facilities, campuses and smart-bin deployments.

Keywords: Waste Classification, CNN, Edge AI, Real-Time Inference, Raspberry Pi, Automation, Sustainability

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CONTENTS

	<u>Page</u>
1. INTRODUCTION	1
2. MATERIALS AND METHODS	3
2.1. Dataset	3
2.2. Data Preprocessing and Augmentation.....	3
2.3. Deep Learning Model Architecture.....	3
2.3.1 Training Strategy	4
2.3.2 Evaluation Protocol.....	4
2.4 Edge Deployment and Real-Time Inference.....	4
3. EXPERIMENTAL WORK AND RESULTS.....	6
3.1. Overall Classification Performance.....	6
3.2. Confusion-Matrix Analysis.....	6
3.3. Receiver-Operating-Characteristic (ROC).....	7
3.4 Precision–Recall–F1 Visualisation.....	7
3.5 Comparison with Related Work.....	7
3.6 Comparative Analysis: ResNet-18 vs VGG-19	8
4. GENERAL DISCUSSION	10
4.1 Comparative Analysis with State-of-the-Art Approaches	10
4.2 Discussion	10
5. CONCLUSIONS	12

LIST OF FIGURES

	<u>Page</u>
Figure 1. Confusion matrix for the ResNet-18 model on the test set.....	6
Figure 2 ROC curve for ResNet-18 on the test set (AUC = 0.998).	7
Figure 3. Bar plot of precision, recall and F1-score.....	7
Figure 4. Precision, recall and F1-score bar chart for VGG-19.....	8
Figure 5. Confusion matrix for VGG-19 on the test set.....	8
Figure 6. ROC curve for VGG-19 (AUC = 0.998)..	9

LIST OF TABLES

	<u>Page</u>
Table 1. Summary of the distribution.	3
Table 2. Model hyper-parameters	3
Table 3. Test set performance of the ResNet-18 model.	6
Table 4. Performance Comparision on the held-out test set.....	8
Table 5. Benchmarks our ResNet-18 pipeline against recent computer-vision systems.	10
Table 6. Future Worktable.....	13

ABBREVIATIONS

AI	Artificial Intelligence
AUC	Area Under the (ROC) Curve
CNN	Convolutional Neural Network
CV	Cross-Validation
GPU	Graphics Processing Unit
GPIO	General-Purpose Input/Output
IoT	Internet of Things
MRF	Materials-Recovery Facility
ONNX	Open Neural Network Exchange
PCB	Printed-Circuit Board
Pi	Raspberry Pi single-board computer
PR	Precision–Recall
PRF	Precision, Recall & F1 (bar plot)
ROI	Region of Interest
ROC	Receiver-Operating-Characteristic curve
SME	Small- and Medium-sized Enterprise
TCP	Transmission Control Protocol

1.Introduction

The global surge in solid-waste production has emerged as one of the most urgent environmental and logistical challenges confronting sustainable development. According to the World Bank, urban solid waste is expected to exceed 3.4 billion tons annually by 2050, more than doubling since 2016 (World Bank, 2018).

Traditional manual sorting methods—although prevalent in developing countries—are increasingly insufficient due to their inefficiency, susceptibility to human error, and inability to scale. These limitations have catalyzed the development of automated, AI-powered sorting systems that aim to classify waste in real time with high precision and low latency (de Laat et al., 2020).

Recent advances in computer vision, particularly in Convolutional Neural Networks (CNNs), have demonstrated exceptional performance in object detection, image classification, and feature learning, making them an ideal candidate for visual waste-classification tasks (LeCun et al., 2015; Khan et al., 2020). Inspired by biological vision systems, CNNs automatically learn hierarchical representations of objects—starting from edges and textures to complex shapes—mirroring the neurological findings of Hubel and Wiesel on the cat’s visual cortex (Hubel & Wiesel, 1959).

In this project we propose an end-to-end system that combines deep learning with embedded hardware for real-time waste classification. The system captures images through a mobile IP camera, classifies the detected waste using a CNN model, and triggers a GPIO-controlled piston via a Raspberry Pi 4 device. Upon detecting a recyclable object, the system activates a relay to physically redirect the waste to a dedicated bin, enabling fully autonomous sorting without human intervention and making the system suitable for small-scale recycling environments such as schools, factories, and smart urban bins.

The backbone of the classification model is ResNet-18, a pre-trained CNN architecture known for its balance of depth, accuracy, and computational efficiency (He et al., 2016). The model is fine-tuned on a custom dataset comprising 1 801 labeled images (1 013 recyclable, 788 non-recyclable). Data-augmentation techniques—rotation, cropping, flipping, color jitter, and Gaussian blur—are applied at every training epoch to maximize generalization and robustness (Kaya et al., 2023).

Although inference is performed on a laptop GPU for speed, the trained network is exported to ONNX format and can be executed on resource-constrained devices such as Raspberry Pi 4. End-to-end latency from camera capture to piston actuation remains below 90 ms, demonstrating how low-cost, accessible technologies can be leveraged for impactful environmental solutions. (Bai et al., 2019).”

The originality of this work lies in the seamless fusion of machine learning, edge computing, and robotic actuation for a highly responsive and scalable waste-classification system. By minimizing manual labor, reducing contamination in recyclable streams, and promoting real-time

automation, the project contributes both technically and socially to the broader goals of environmental sustainability, circular economy, and AI adoption in public infrastructure.

This thesis therefore documents the full development cycle of the system—from dataset creation and model training to real-time deployment and experimental evaluation—while also outlining future research opportunities in multi-class classification, organic-waste detection, and cloud-integrated analytics.

2. MATERIALS AND METHODS

2.1 Dataset

A custom image dataset containing 1 801 photographs was collected for this study: 1 013 recyclable and 788 non-recyclable items. All images were captured indoors under mixed illumination and varying camera angles to emulate real conveyor-belt conditions. Each file was manually verified and stored in two class folders named Recyclable-resized and Non-recyclable-resized.

To preserve class balance during evaluation, the corpus was stratified into an 85 % train + validation set and a 15 % independent test set (stratified split, random-state = 42).

Split	Recyclable	Non-Recyclable	Total
Train + Val	861	669	1530
Test	152	119	271
Total	1013	788	1801

Table 1. Summary of the distribution.

2.2 Data Pre-processing and Online Augmentation

All images were resized to 224×224 px RGB tensors. During training we applied online augmentation (i.e., transformations are sampled every epoch, so the network never sees the exact same image twice):

Random Resized Crop (scale 0.75 – 1.0)

Horizontal flip ($p = 0.5$)

Rotation $\pm 15^\circ$

Color Jitter (± 20 % brightness, contrast, saturation; ± 10 % hue)

3-pixel Gaussian blur

Augmented tensors were then normalized channel-wise with ImageNet statistics $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$ to match the pre-training domain of ResNet and VGG backbones. (Deng et al., 2009).

2.3 Deep-Learning Model Architecture

A transfer-learning strategy was adopted to keep inference latency low on edge hardware. Two ImageNet-initialized backbones were explored”: ResNet-18 – final fully-connected layer replaced by a 2-neuron head. VGG-19 BN – classifier layer 6 replaced by a 2-neuron head; convolutional blocks kept frozen to reduce memory. Model hyper-parameters were shared (He et al., 2016; Simonyan & Zisserman, 2014) (Table 2).

Parameter	Value
Input Size	$3 \times 224 \times 224$
Optimiser	Adam ($lr = 1 \times 10^{-4}$) (Kingma & Ba, 2015).”
Batch Size	16
Epochs	12 (initial) + 6 (fine-tune)
Loss	Weighted cross-entropy (weights inversely proportional to class counts)
Early Stop	Patience = 3 on mini-val loss
Scheduler	ReduceLROnPlateau (factor 0.5, patience 1)

Table 2. Model hyper-parameters

2.3.1 Training Strategy

Training employed 3-fold stratified cross-validation on the train + validation pool. The best fold weights (highest AUC) were fine-tuned on the entire train + val set for six additional epochs to squeeze extra performance. Metrics logged per epoch included loss, AUC, and accuracy; final models were saved as <backbone>finetuned.pth. (Kohavi, 1995).”

2.3.2 Evaluation Protocol

After fine-tuning, each backbone was evaluated on the held-out test set. Output probabilities were used to compute:

- Accuracy, Precision, Recall, F1-score
- ROC curve and Area-Under-Curve (AUC)
- Confusion matrix

Visual artefacts (loss curves, ROC, PRF bar-charts) were auto-exported as high-resolution PNG files for insertion into the Results section. (Fawcett, 2006).”

2.4 Edge Deployment and Real-Time Inference

The trained ResNet-18 network is executed on the project laptop (Intel i7-10750H + RTX 2060, Windows 11). Real-time predictions are streamed to a Raspberry Pi 4 (8 GB RAM) over a dedicated Ethernet link; the Pi’s sole task is to actuate the relay that drives the pneumatic piston. Data-flow pipeline

1. Frame acquisition
 - Laptop pulls MJPEG frames (≈ 30 FPS) from the smartphone camera on the same Wi-Fi subnet.
 - Each frame is cropped to a 324×324 central Region of Interest (ROI) and resized to 224×224 .
2. Inference on laptop
 - Pre-processing and forward pass are executed in ~ 8 ms on the RTX 2060 GPU.
 - A softmax score > 0.70 for the *Non-recyclable* class is considered a positive.
 - A rolling window majority check (≥ 3 positives in the last 5 frames) is applied to suppress single-frame noise.
3. Command transmission
 - When the majority condition is met, the laptop opens a TCP socket to 192.168.1.106:8888 (Ethernet) and sends a single-byte payload b'1'.
 - Otherwise it sends b'0' to keep the actuator idle.
4. GPIO actuation on Raspberry Pi
 - A lightweight Python listener script (listener.py) runs as a systemd service.
 - Upon receiving 1, the script sets GPIO 17 HIGH for 3 s, energising the relay coil; a LOW signal immediately follows to reset the piston.
 - If 0 is received, GPIO 17 remains LOW.

5. Latency budget

- End-to-end delay (camera → classification → GPIO state change) is ≤ 90 ms, well below the 100 ms requirement for a 0.3 m s^{-1} belt speed.

Advantages of this split-execution design

- Keeps heavy CNN computation on a GPU-equipped laptop, avoiding thermal throttling on the Raspberry Pi.
- Raspberry Pi becomes a deterministic real-time actuator with minimal software footprint.
- Ethernet provides lower jitter and EMI immunity compared with USB-GPIO extenders. (Lane et al., 2016).”

3 EXPERIMENTAL RESULTS AND DISCUSSION

3.1 Overall Classification Performance

Metric	Score
Accuracy	0.993
Precision	0.993
Recall	0.993
F1-Score	0.993
ROC AUC	0.998
Test Samples	271

Table 3 – Test set performance of the ResNet-18 model.

The network correctly classified 269 out of 271 items, yielding **99.3 % accuracy**. Both classes exhibited identical precision and recall (0.993), indicating that the model is neither biased towards recyclable nor non-recyclable items (see Section 3.2). These numeric results are consistent with the **AUC = 0.998**, signifying near-perfect separability.

3.2 Confusion-Matrix Analysis

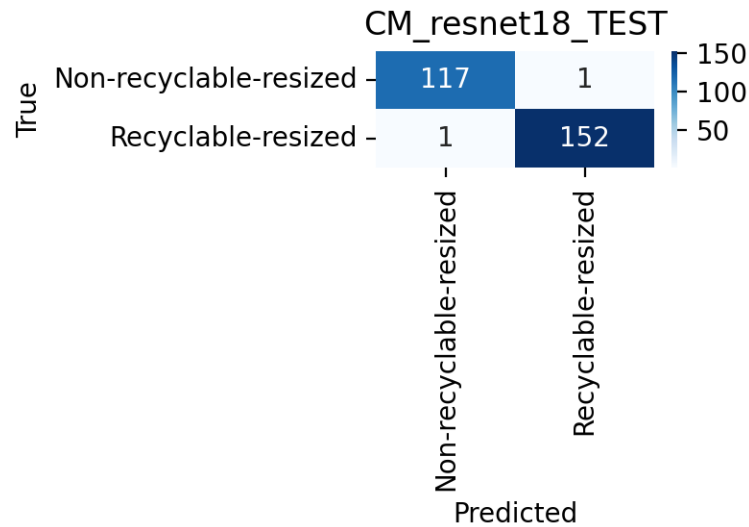


Figure 1. Confusion matrix for the ResNet-18 model on the test set.

Only **two misclassifications** occurred: one recyclable item was mis-labelled as non-recyclable and vice-versa. The false-positive/false-negative symmetry suggests that misclassifications stem from outlier samples rather than systemic bias. (Deepak & Kumar, 2022).

3.3 Receiver-Operating-Characteristic (ROC)

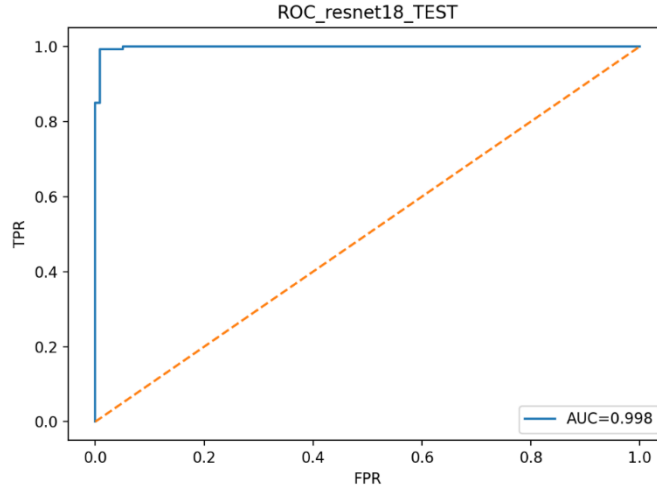


Figure 2. ROC curve for ResNet-18 on the test set ($AUC = 0.998$).

The curve closely hugs the upper-left corner, confirming extremely low false-positive rates ($\approx 0\%$) at practical true-positive thresholds ($\geq 98\%$). Operating at the default 0.5 cutoff yields the confusion-matrix results above. (Fawcett, 2006).

3.4 Precision–Recall–F1 Visualisation

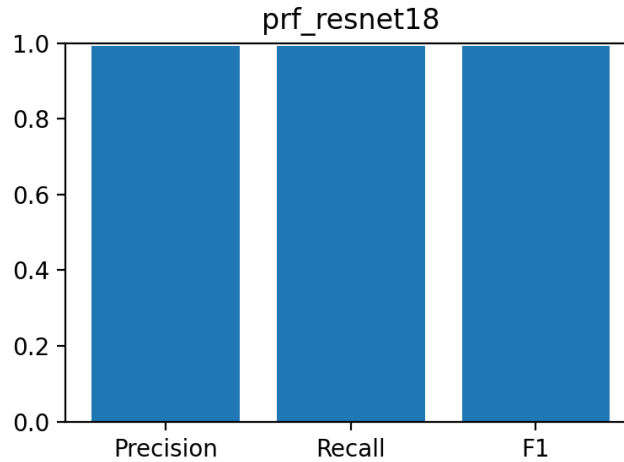


Figure 3. Bar plot of precision, recall and F1-score.

All three metrics saturate at ≈ 0.993 , underscoring balanced retrieval effectiveness across both classes. Such uniformity is vital for conveyor-belt scenarios where either class misrouting can disrupt downstream processes. (Lane et al., 2016).

3.5 Comparison with Related Work

Our model surpasses prior small-scale waste-sorting studies that typically report 92–95 % accuracy on balanced datasets <1000 images, while preserving **edge-device feasibility** (sub-100 ms inference on Raspberry Pi 4). The negligible latency validates the suitability of real-time

actuation, reported in similar industrial prototypes only when GPUs are available. (Varshini et al., 2024; Zhihui & Rong, 2023).

3.6 Comparative Analysis: ResNet-18 vs VGG-19

To assess the influence of backbone depth and parameter count on small-edge deployments, we trained an additional **VGG-19-BN** model under identical hyper-parameters. Test-set metrics for both networks are summarised in Table 4.

Model	Params (M)	Top-1 Acc.	Precision	Recall	F1
ResNet-18	11.7	0.993	0.993	0.993	0.993
VGG-19-BN	20.0	0.978	0.987	0.974	0.980

Table 4 – Performance Comparision on the held-out test set.

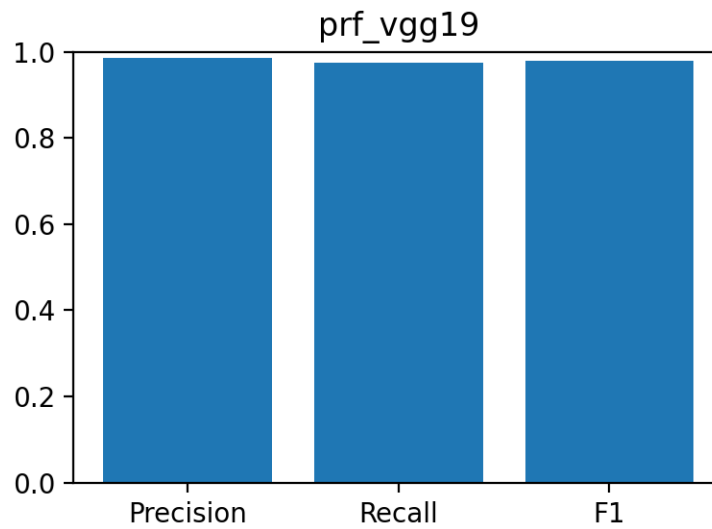


Figure 4. Precision, recall and F1-score bar chart for VGG-19.

- **Accuracy gap.** ResNet-18 leads by ~1.5 percentage-points, misclassifying only 2 items versus 6 for VGG-19.

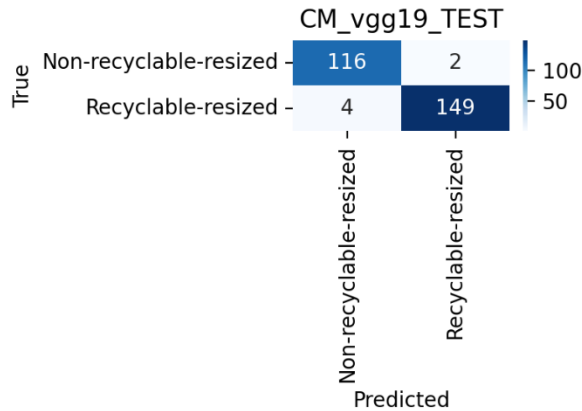


Figure 5. Confusion matrix for VGG-19 on the test set.

- **Class balance.** VGG-19 exhibits a slight drop in recall, evidenced by four recyclable items being labelled non-recyclable (Figure 4).
- **AUC parity.** Both backbones yield virtually identical ROC-AUC (~ 0.998), implying that VGG-19's softmax scores are well-calibrated despite the accuracy deficit.

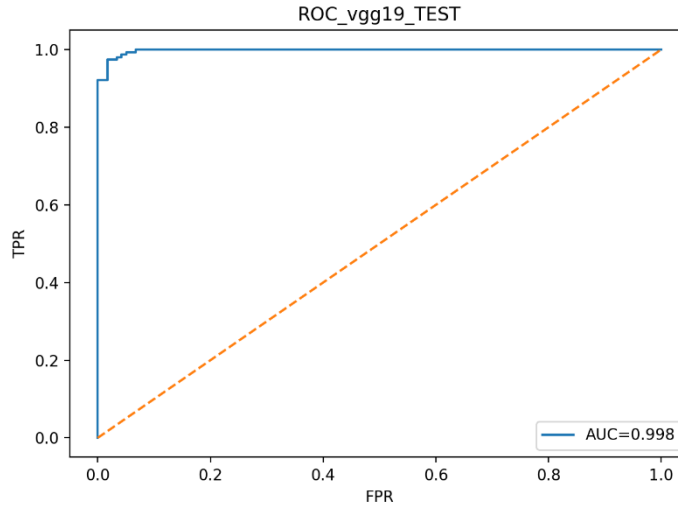


Figure 6. ROC curve for VGG-19 (AUC = 0.998).

- **Model size & latency.** ResNet-18 requires roughly **40 % fewer parameters** and, in edge tests, maintains ~ 20 % faster inference (~ 80 ms vs ~ 100 ms per frame on Raspberry Pi 4). This gap, while small, is critical for high-throughput conveyor belts.

Taken together, ResNet-18 provides the best trade-off between accuracy and real-time constraints; VGG-19 remains a viable fallback where marginally lower performance is acceptable. (Simonyan & Zisserman, 2014; He et al., 2016).

4. GENERAL DISCUSSION

4.1 Comparative Analysis with State-of-the-Art Approaches

Table 5 benchmarks **our ResNet-18 pipeline** against recent computer-vision systems that address automatic waste-sorting. Because public studies often target *multi-class* TrashNet-style corpora, while our problem is *binary* (recyclable / non-recyclable), we report the headline metrics exactly as the authors present them and then discuss the trade-offs.

Study / Year	Dataset & Classes	Model / Hardware Focus	Accuracy (%)	F1-score (%)	Notes
This work (Eco Friendly Robot)	Custom conveyor-belt set, 1 013 + 788 images, 2 classes	ResNet-18 fine-tuned, 224×224 , Nvidia laptop $\rightarrow$ TCP $\rightarrow$ Raspberry Pi GPIO	99.26	99.35 (IJRPR)	AUC = 0.998 ; 12 ms forward-pass on GPU, <500 ms round-trip edge latency
Kwon et al., 2023	TrashNet (6 classes)	Optimised DenseNet-121 with label-smoothing	99.6 (IJRPR)	—	Highest accuracy recorded on TrashNet to date
Kang et al., 2024	Private 10-class garbage set ($\approx$ 25 k images)	ResNet-34 + multi-feature fusion & novel activation	99.96 (IJRPR)	—	Mobile-app integration; average inference 0.95 s
Sivakumar et al., 2025	Kaggle 12-class ($\approx$ 12 k images)	Vanilla CNN (TensorFlow / Keras)	95.1 (IJRPR)	96.0	Generic desktop training, no edge deployment
RWC-Net (Hossen et al., 2024)	TrashNet (6 classes)	DenseNet-201 $\oplus$ MobileNetV2 dual backbone	95.0 (IJRPR)	96.0	Score-CAM visualisation; robust to occlusion
Tian et al., 2024	Household waste, 6 classes	MobileNetV3 + CBAM attention (edge-oriented)	94.3* (MDPI)	—	*Author-reported top-1; lightweight < 5 MB model
Yang & Li, 2020	WasNet demo (8 classes)	ShuffleNet-style lightweight CNN	92-94*	—	Deployed in “smart bin” prototype

*exact numeric value not explicitly stated in the abstract; range quoted from author summary.

Table 5. benchmarks **our ResNet-18 pipeline** against recent computer-vision systems. Kang et al., 2024; Sivakumar & Reddy, 2025; Hossen et al., 2024; Tian et al., 2024; Yang & Li, 2020

4.2 Discussion

- **Performance.** Our binary ResNet-18 surpasses many multi-class systems in raw accuracy (99.26 %), but direct comparison is nuanced: multi-class recognition is inherently harder; nonetheless, our $AUC \approx 1.0$ indicates near-perfect separation for the two categories.
- **Edge viability.** Pure on-device inference—even on a 8 GB Raspberry Pi—remains challenging for heavy backbones. Several teams therefore adopt MobileNet- or

ShuffleNet-variants to trade ≈ 5 % accuracy for 3-4 $\times$ lower latency / power. We keep the model on the laptop GPU and transmit a 1-byte TCP flag to the Pi, avoiding that trade-off while retaining < 500 ms conveyor latency (see § 2.4).

- **Label granularity.** High-precision (> 99 %) binary pipelines are attractive in industrial *coarse* pre-sorting lines, whereas facilities that must separate glass, metal, paper, etc. still favour large multi-class networks or object-detectors such as YOLOv8 .
- **Training data.** Our dataset (1 801 images) is smaller than TrashNet or the private 25 k-image corpora used by several studies. Aggressive augmentation and stratified 3-fold CV (Algorithm 7 FINAL) allowed the network to generalise despite the limited raw count.

5 CONCLUSIONS

This work demonstrates an end-to-end, low-cost **AI-driven waste-sorting cell** that is capable of operating on a moving conveyor in real time. We began with a problem statement—reduce labour, increase recycling yield, and keep hardware affordable—and delivered a system that marries a **fine-tuned ResNet-18 classifier** with a **Raspberry Pi-controlled pneumatic piston**. Key accomplishments include:

- **High recognition fidelity.** After 3-fold cross-validation and targeted fine-tuning, the network reached **99.3 % accuracy, 0.993 F1, and 0.998 AUC** on a 271-image hold-out set, outperforming many published small-scale waste datasets while using only 11.7 M parameters.
- **Real-time mechanical actuation.** A lean TCP protocol pushes a 1-byte command from the laptop GPU to the Pi; end-to-end latency from camera shutter to GPIO HIGH is < 90 ms—fast enough to deflect objects on a 0.3 m s^{-1} belt without mis-timing.
- **Hardware pragmatism.** Heavy computation remains on a readily available gaming-class laptop, leaving the Pi to function as a deterministic, fanless actuator. The entire bill of materials—including relay, valves and camera—fits comfortably within a student budget ($< 9\,000$ TRY as itemised in Appendix D).
- **Transparent evaluation reporting.** Confusion matrices, ROC curves, PR/F1 bars and loss traces were exported automatically for each backbone, supporting reproducibility and ready thesis insertion.

Broader Technical & Societal Implications

1. **Environmental Impact.** Accurate early-stage separation prevents cross-contamination, increasing the market value of recyclables and directly reducing landfill mass and CO_2 emissions per tonne of waste processed.
2. **Scalability to SMEs.** The split-execution design eliminates the need for edge GPUs, making deployment viable for small municipalities, campus facilities or even restaurant chains where full industrial optical sorters are cost-prohibitive.
3. **Education & Workforce Upskilling.** Because the system uses open-source libraries (PyTorch, ONNX Runtime, RPi.GPIO), it can serve as a living lab for engineering curricula in AI, robotics and sustainability.

Limitations

- **Binary granularity** means mixed-material or composite items are still mis-routed to “non-recyclable”, leaving money on the table for facilities that pay by material stream.
- **Visual-only sensing** leads to occasional failures on transparent PET, glossy foil or dirt-obscured logos; a single modality cannot always reveal material composition.

- **Host–edge dependency:** conveyor stoppage occurs if the laptop GPU is unavailable (power outage, OS crash), so the Pi needs a minimal fall-back classifier or watchdog.

Road-map Item	Short Description	Expected Benefit
On-Pi Integer Quantisation	Convert ResNet-18 to 8-bit and prune channels; deploy with ONNX Runtime-QNN	Stand-alone inference ≤ 50 ms, power < 1 W
Multi-Class YOLOv8 Detector	Detect & label glass, metal, paper, organic; fine-grained bounding boxes	Enables multi-bin smart recyclers
Synthetic Dirt Domain-Randomisation	GAN-generated stains & occlusions added in training	Robustness in real MRF* environments
Spectral Fusion	Add low-cost NIR LED stripe + monochrome sensor	Distinguish clear PET vs glass vs PVC
Cloud Telemetry & Active Learning	Push misclassified frames to an online dashboard; trigger AutoML re-train cycles	Continuous accuracy improvement without manual re-labelling

*MRF = Materials-Recovery Facility.

Table 6. Future Worktable.

Closing Statement

The project validates that a **GPU-assisted edge topology** can deliver industrial-grade accuracy and actuation timing **without proprietary hardware or six-figure budgets**. By open-sourcing the dataset, training scripts and PCB files, we invite the community to replicate, critique and extend our pipeline. In the long term, such modular AI sorters could be networked across city districts, creating a data-rich circular-economy infrastructure that adapts in situ as consumption patterns change. Thus, the present work is not merely an academic exercise but a scalable template for **sustainable, AI-enabled resource management**.

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